Version 7.4.8

PVsyst - Simulation report

Grid-Connected System

Project: 221208 - Lundby

Variant: 251014- Lundby - PVsyst report- Grid Studies-Final run- 66% 33% 720 725

Ground system (tables) on a hill

System power: 8594 kWp

Grunnarp - Sweden

Author

Svea Solar Utility Services AB (Sweden)

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Svea Solar Utility Services AB (Sweden)

Project summary

Geographical Site
Grunnarp
Sweden

Situation

Latitude	57.15 °N
Longitude	12.34 °E
Altitude	31 m
Time zone	UTC+1

Weather data
Grunnarp
Solcast TMY

Monthly albedo values

	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
Albedo	0.30	0.30	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.30

System summary

Grid-Connected System

Ground system (tables) on a hill

PV Field Orientation
Fixed plane
Tilt/Azimuth 26 / 0 °

Near Shadings
According to strings : Slow (simul.)
Electrical effect 80 %

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	11908 units
Pnom total	8594 kWp

Inverters

Nb. of units	21 units
Pnom total	6720 kWac
Grid power limit	6500 kWac
Grid lim. Pnom ratio	1.322

Results summary

Produced Energy	8700535 kWh/year	Specific production	1012 kWh/kWp/year	Perf. Ratio PR	85.22 %

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General parameters

Grid-Connected System | **Ground system (tables) on a hill**

PV Field Orientation

Orientation
Fixed plane
Tilt/Azimuth 26 / 0 °

Sheds configuration
Nb. of sheds 265 units
Identical arrays
Sizes
Sheds spacing 7.71 m
Collector width 4.79 m
Ground Cov. Ratio (GCR) 62.2 %
Shading limit angle
Limit profile angle 31.7 °

Models used
Transposition Perez
Diffuse Imported
Circumsolar separate

Horizon
Average Height 3.7 °

Near Shadings
According to strings : Slow (simul.)
Electrical effect 80 %

User's needs
Unlimited load (grid)

Bifacial system
Model 2D Calculation unlimited sheds

Bifacial model geometry

Sheds spacing	7.71 m
Sheds width	4.79 m
Limit profile angle	31.7 °
GCR	62.2 %
Height above ground	0.80 m

Bifacial model definitions

Ground albedo average	0.22
Bifaciality factor	80 %
Rear shading factor	5.0 %
Rear mismatch loss	10.0 %
Shed transparent fraction	0.0 %

Monthly ground albedo values

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Year
0.30	0.30	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.30	0.22

Grid power limitation
Active power 6500 kWac
Pnom ratio 1.322

PV Array Characteristics

PV module
Manufacturer Tongwei Co., Ltd.
Model TWMNF-66HD720
(Custom parameters definition)
Unit Nom. Power 720 Wp
Number of PV modules 7930 units
Nominal (STC) 5710 kWp

Inverter
Manufacturer Sungrow
Model SG350HX-20
(Custom parameters definition)
Unit Nom. Power 320 kWac
Number of inverters 14 units
Total power 4480 kWac

Array #1 - Transformer 1
Number of PV modules 4004 units
Nominal (STC) 2883 kWp
Modules 154 string x 26 In series
At operating cond. (50°C)
Pmpp 2683 kWp
U mpp 986 V
I mpp 2720 A

Number of inverters 7 units
Total power 2240 kWac
Operating voltage 500-1500 V
Max. power (=>30°C) 352 kWac
Pnom ratio (DC:AC) 1.29
Power sharing within this inverter

PV Array Characteristics

Array #2 - Transformer 2

Number of PV modules	3926 units	Number of inverters	7 units
Nominal (STC)	2827 kWp	Total power	2240 kWac
Modules	151 string x 26 In series		
At operating cond. (50°C)		Operating voltage	500-1500 V
Pmpp	2630 kWp	Max. power (=>30°C)	352 kWac
U mpp	986 V	Pnom ratio (DC:AC)	1.26
I mpp	2667 A	Power sharing within this inverter	

Array #3 - Transformer 3

PV module		**Inverter**	
Manufacturer	Tongwei Co., Ltd.	Manufacturer	Sungrow
Model	TWMNF-66HD725	Model	SG350HX-20
(Custom parameters definition)		(Custom parameters definition)	
Unit Nom. Power	725 Wp	Unit Nom. Power	320 kWac
Number of PV modules	3978 units	Number of inverters	7 units
Nominal (STC)	2884 kWp	Total power	2240 kWac
Modules	153 string x 26 In series	Operating voltage	500-1500 V
At operating cond. (50°C)		Max. power (=>30°C)	352 kWac
Pmpp	2683 kWp	Pnom ratio (DC:AC)	1.29
U mpp	990 V	Power sharing within this inverter	
I mpp	2710 A		

Total PV power		**Total inverter power**	
Nominal (STC)	8594 kWp	Total power	6720 kWac
Total	11908 modules	Max. power	7392 kWac
Module area	36990 m²	Number of inverters	21 units
		Pnom ratio	1.28

Array losses

Array Soiling Losses

Average loss Fraction 5.0 %

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
18.0%	11.0%	4.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	2.0%	18.0%

Thermal Loss factor

Module temperature according to irradiance

Uc (const)	29.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	1.1 mΩ
Global wiring resistance	0.38 mΩ
Loss Fraction	0.3 % at STC

Serie Diode Loss

Voltage drop	0.7 V
Loss Fraction	0.1 % at STC

LID - Light Induced Degradation

Loss Fraction 1.0 %

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction 0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

System losses

Auxiliaries loss

Proportional to Power	7.5 W/kW
0.0 kW from Power thresh.	
Night aux. cons.	200 W

AC wiring losses

Inv. output line up to MV transfo

Inverter voltage	800 Vac tri
Loss Fraction	1.24 % at STC
Inverter: SG350HX-20	
Wire section (21 Inv.)	Alu 21 x 3 x 240 mm²
Average wires length	150 m

MV line up to Injection

MV Voltage	10.5 kV
Average loss Fraction	0.16 % at STC

Array #1 - Transformer 1		**Array #2 - Transformer 2**	
Wires	Alu 3 x 300 mm²	Wires	Alu 3 x 300 mm²
Length	5 m	Length	608 m

Array #3 - Transformer 3	
Wires	Alu 3 x 300 mm²
Length	1184 m

AC losses in transformers

MV transfo

Medium voltage	10.5 kV
3 identical MV transfos	

Transformer from Datasheets		**Operating losses at STC (full system)**	
Nominal power	2500 kVA	Nb. identical MV transfos	3
Iron Loss	1.58 kVA	Nominal power at STC	2.84 MVA
Iron loss fraction	0.06 % of PNom	Iron loss	1.58 kVA
Copper loss	18.50 kVA	Iron loss fraction	0.06 % at STC
Copper loss fraction	0.74 % at PNom	Copper loss	23.81 kVA
Coils equivalent resistance	3 x 1.89 mΩ	Copper loss fraction	0.84 % at STC

Horizon definition

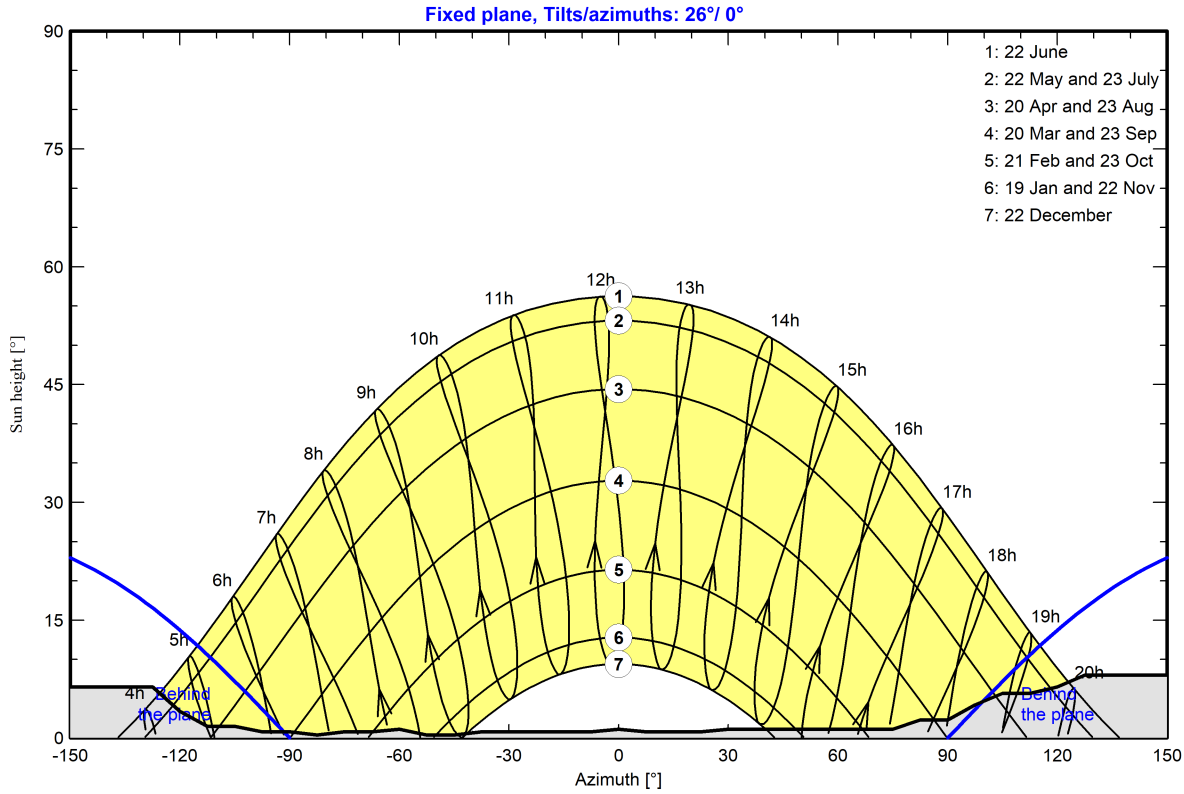
Horizon from PVGIS website API, Lat=57°9'7", Long=12°20'18", Alt=31m

Average Height	3.7 °	Albedo Factor	0.96
Diffuse Factor	1.00	Albedo Fraction	100 %

Horizon profile

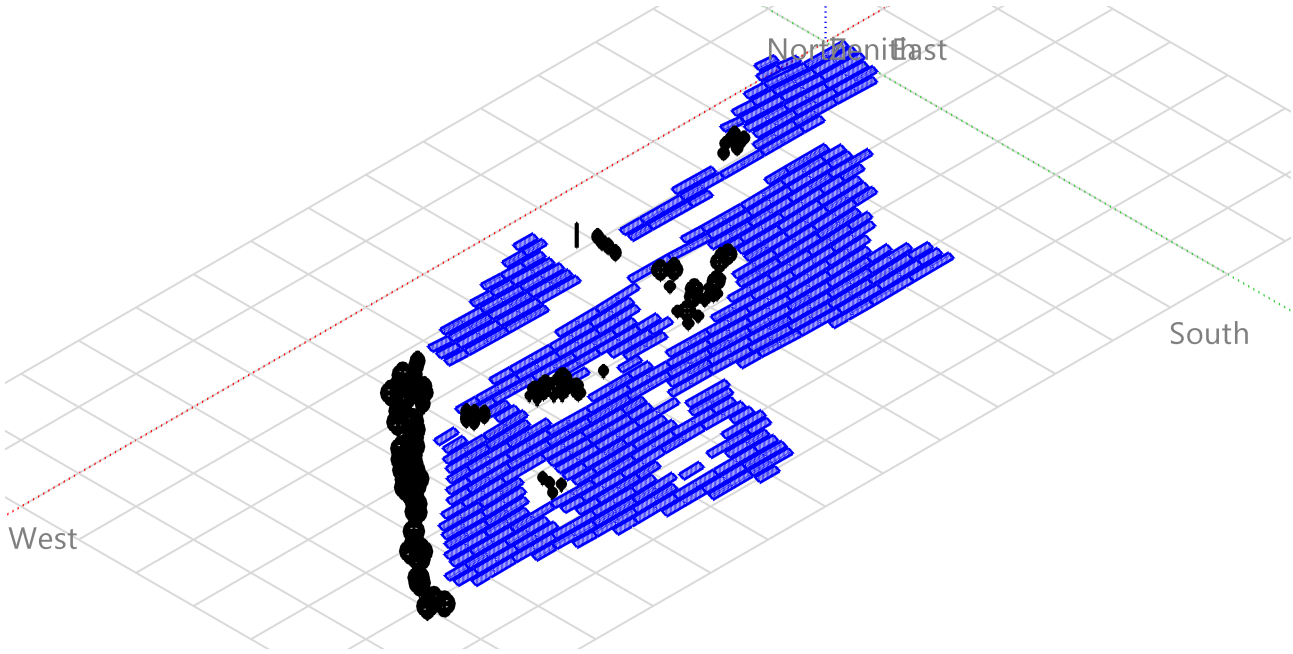
Azimuth [°]	-180	-158	-150	-128	-120	-113	-105	-98	-90	-83	-75
Height [°]	9.2	9.2	6.5	6.5	3.4	1.5	1.5	0.8	0.8	0.4	0.8
Azimuth [°]	-68	-60	-53	-45	-38	-8	0	8	23	30	75
Height [°]	0.8	1.1	0.4	0.4	0.8	0.8	1.1	0.8	0.8	1.1	1.1
Azimuth [°]	83	90	98	105	113	120	128	150	158	180	
Height [°]	2.3	2.3	4.2	5.7	5.7	6.5	8.0	8.0	9.2	9.2	

Sun Paths (Height / Azimuth diagram)



Near shadings parameter

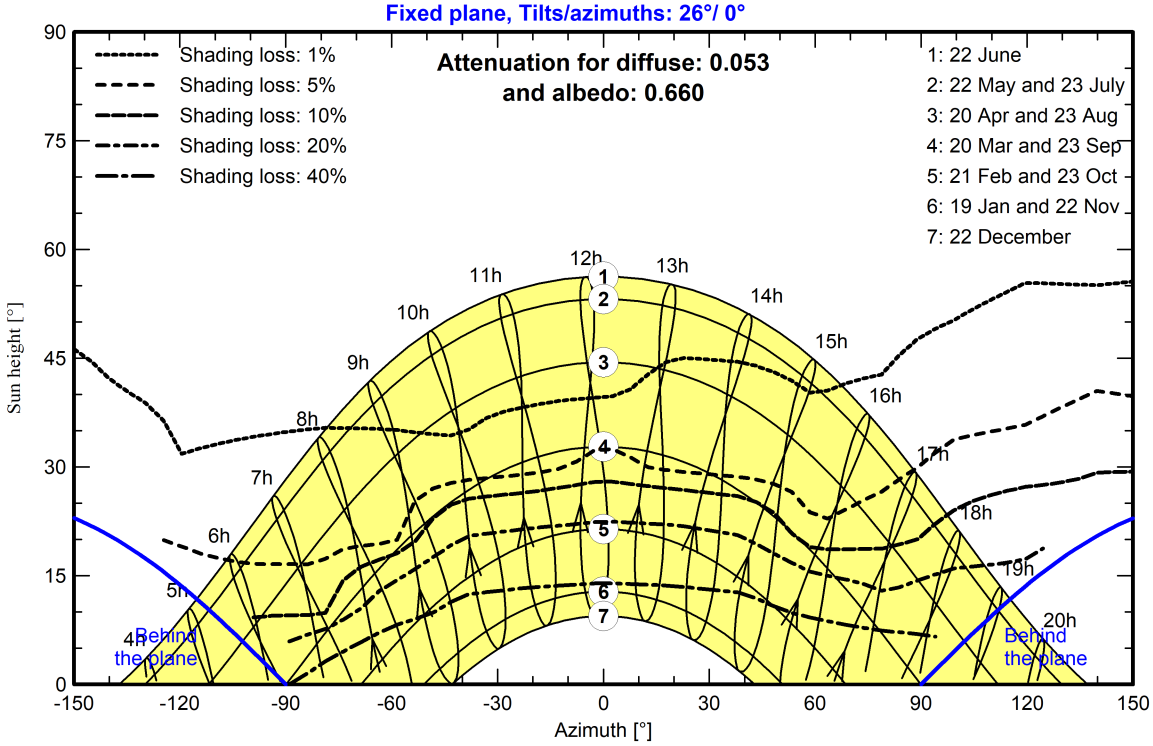
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 26°/ 0°



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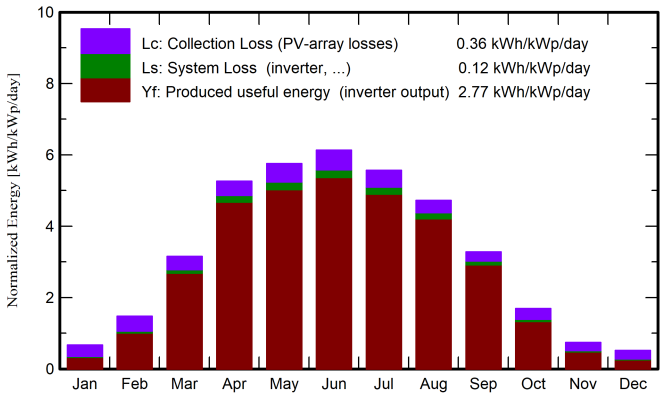
Svea Solar Utility Services AB (Sweden)

Main results

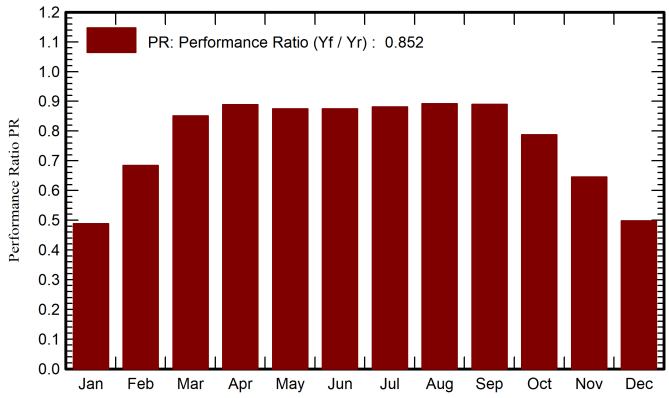
System Production

Produced Energy	8700535 kWh/year	Specific production	1012 kWh/kWp/year
		Perf. Ratio PR	85.22 %

Normalized productions (per installed kWp)



Performance Ratio PR



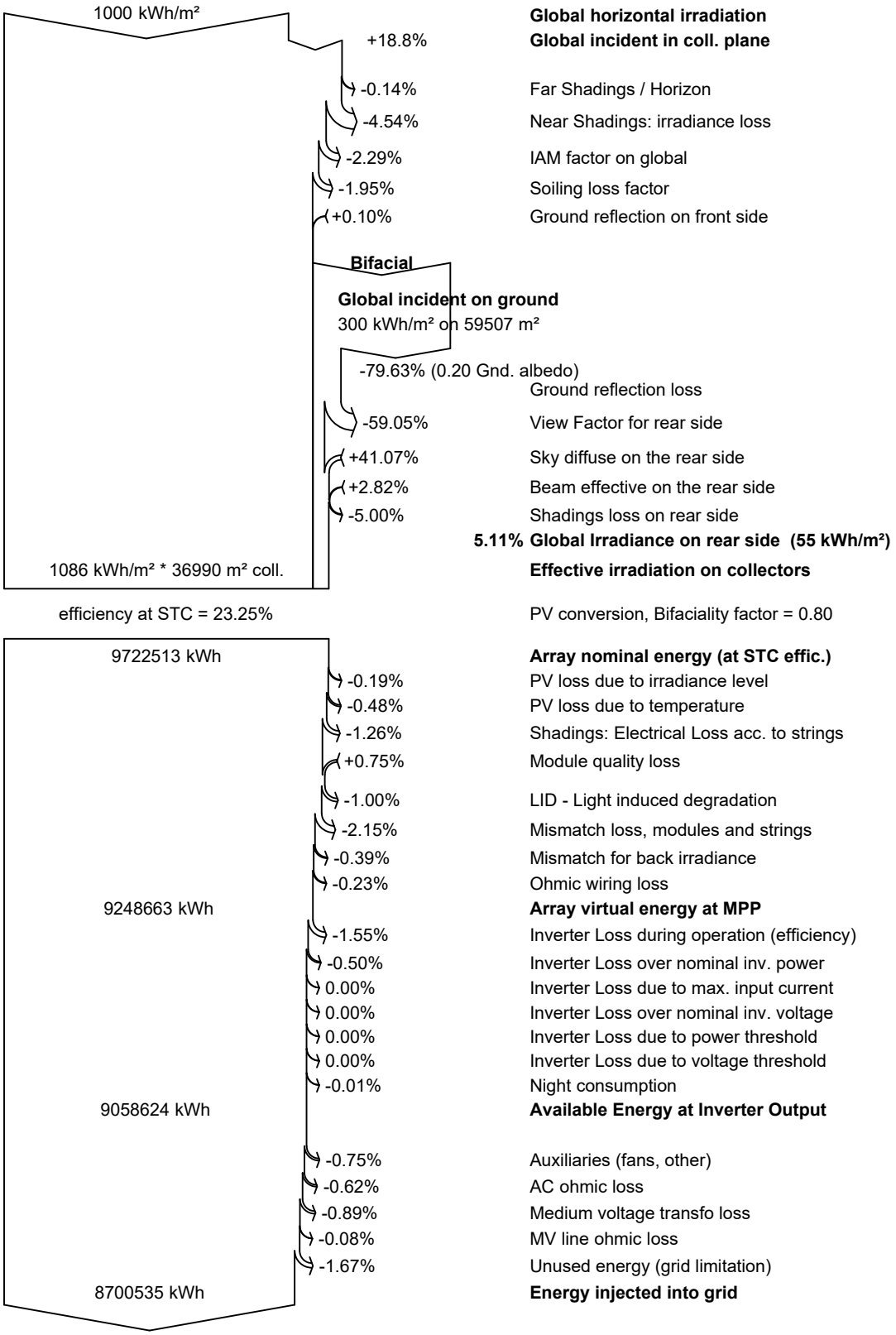
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	10.8	7.73	0.37	20.7	11.5	95156	86714	0.488
February	25.9	15.45	3.99	41.4	30.6	254987	243375	0.684
March	69.2	34.48	1.11	97.8	87.2	742406	715141	0.851
April	129.3	49.30	7.24	157.8	149.5	1255396	1205537	0.889
May	161.7	60.70	11.86	178.4	169.0	1396241	1340415	0.874
June	175.1	67.18	16.52	184.0	174.2	1439760	1383155	0.875
July	163.5	72.30	15.89	172.5	162.7	1358403	1305442	0.881
August	128.0	62.31	16.67	146.4	138.3	1167089	1122236	0.892
September	78.1	41.29	15.60	98.3	92.1	780978	752029	0.890
October	35.8	20.69	10.04	52.4	45.5	370883	354966	0.788
November	13.5	9.40	6.78	22.2	16.5	132849	123247	0.645
December	8.8	7.11	-1.91	16.0	8.9	75827	68279	0.498
Year	999.8	447.93	8.69	1188.0	1086.1	9069974	8700535	0.852

Legends

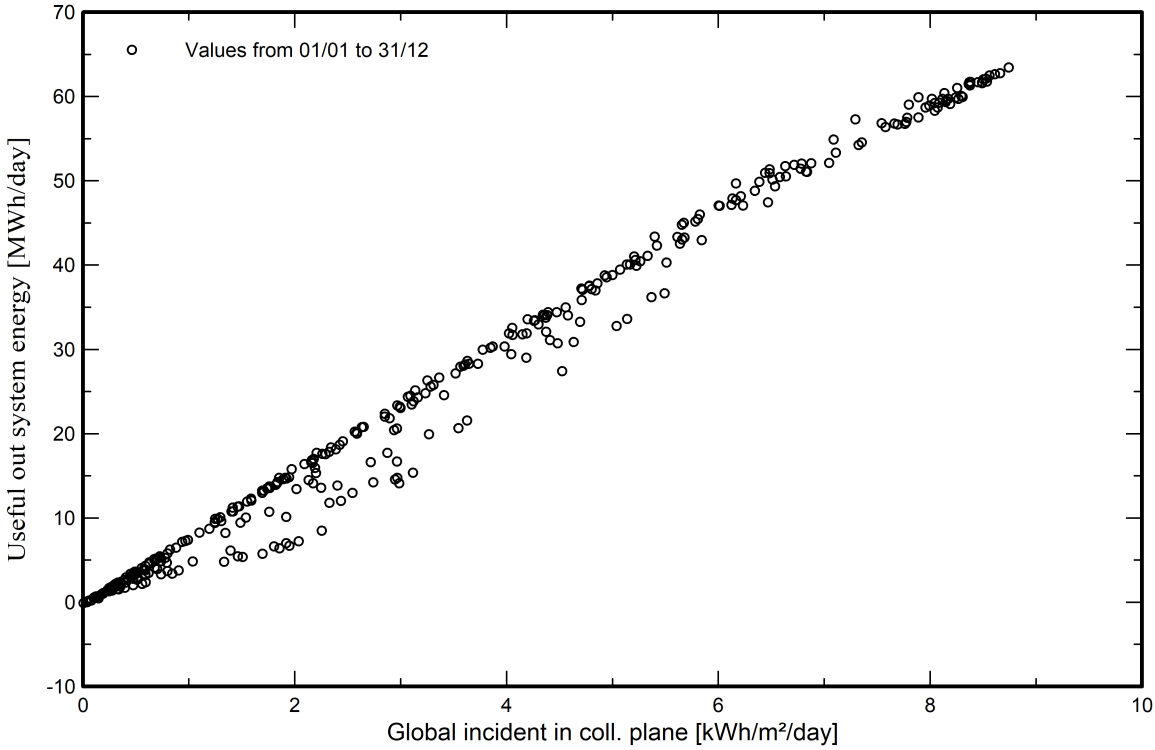
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram

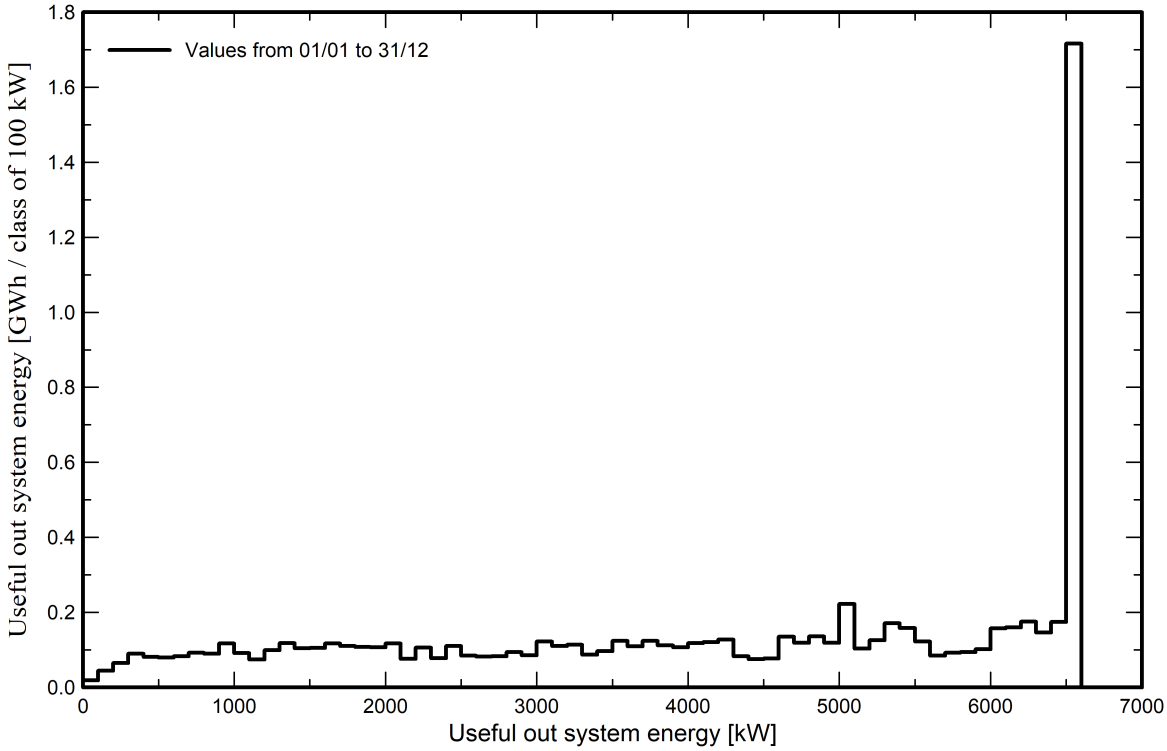


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Solcast TMY
Kind	TMY, multi-year
Year-to-year variability(Variance)	7.3 %
Specified Deviation	
Climate change	0.0 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

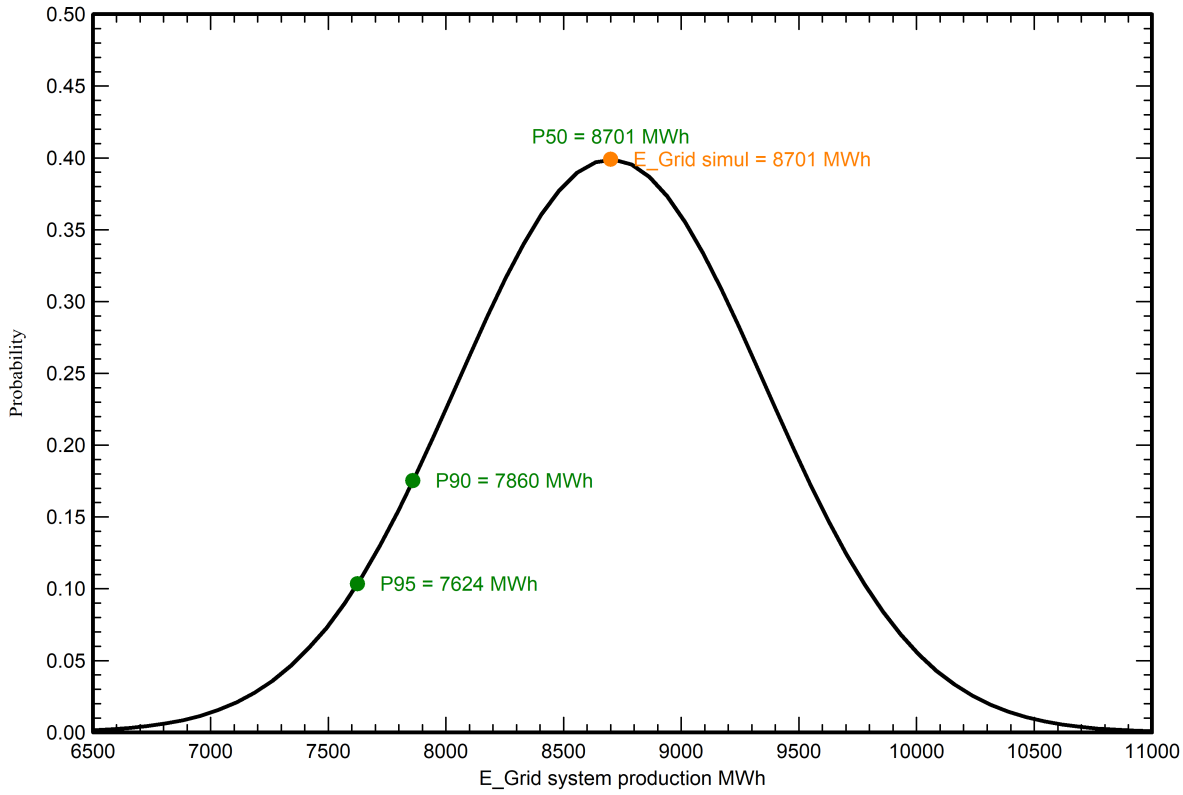
Global variability (weather data + system)

Variability (Quadratic sum)	7.5 %

Annual production probability

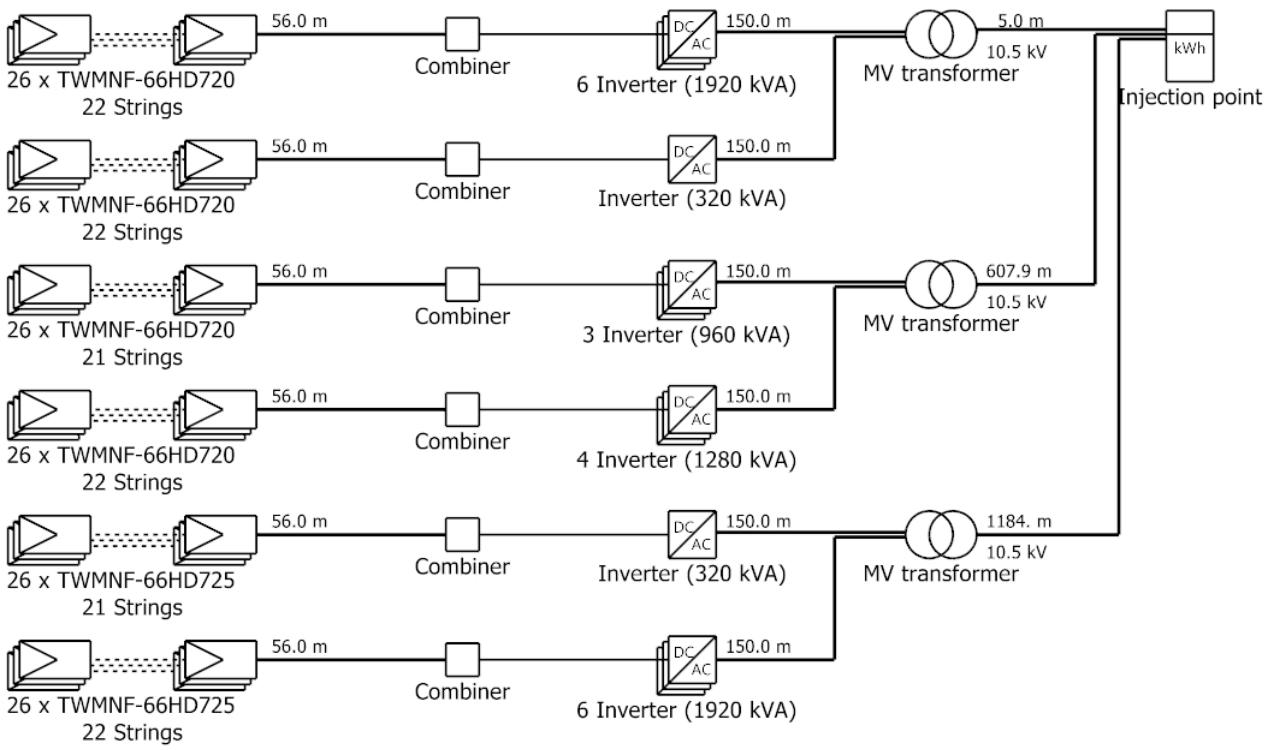
Variability	655 MWh
P50	8701 MWh
P90	7860 MWh
P95	7624 MWh

Probability distribution



Single-line diagram

PVsyst V7.4.8
VD1, Simulation date:
07/11/25 15:22
with V7.4.8



PV module 1	TWMNF-66HD720
PV module 2	TWMNF-66HD725
Inverter	SG350HX-20
String 1	26 x TWMNF-66HD720
String 2	26 x TWMNF-66HD725

221208 - Lundby	Svea Solar Utility S ervices AB (Swede
VD1 : 251014- Lundby - PVsyst report- Grid Studies-Final run- 66% 33% 720	07/11/25